1 The nature of technology: Computer networks

You will learn

- how digital devices can be connected to make networks
- what the internet is and what services it provides
- how the internet helps us work together in the modern world.



A computer is a powerful tool that we use for working and learning. Connecting computers together makes them more powerful and more useful. When computers are connected, we can use them to communicate with each other. We can share files and search for information on the web. We use networks to connect computers together. In this unit you will learn how networks improve the way we live our lives.



Talk about...

Do you think you will use computers in your job when you leave school? Would you like to work in computing when you leave school? Can you think of any jobs that don't use computers?



Learning outcomes: Explain that digital devices can be connected by communication links; Explain what the internet is and some internet services such as the world wide web; Describe some ways the internet helps us work together in the modern world

network network device
local area network
wide area network
internet world wide web
Wi-Fi server hub router



Class activity

Here are two statements about how young people should use the internet.

1 "The internet is dangerous. Young people should not be allowed to use the internet."

2 "Young people should be able to use the internet whenever and wherever they choose."

Discuss the statements in your class. List the arguments for and against each statement.

Can your class write your own statement about how young people should use the internet?



Digital citizen of the future

Whatever you do in your future life, computers and networks will play an important part. The world of computing changes quickly. You will need to keep your skills up to date so you can use technology at work and at home. At work you will go on training courses to learn new skills. At home you will use the internet to learn new skills. As a good digital citizen, you will help people who have fallen behind with their computing skills.



Did you know?

Today, almost every computer is connected to a network. The biggest network in the world is the internet. The internet has the potential to connect computers wherever they are.



1.1 What is a network?

In this lesson

You will learn:

- what a computer network is
- why networks are important at school, at home and in the workplace.

Spiral back

Last year, you learned about how computers help us learn, work and enjoy our free time. Now you will learn about how computers and other devices are connected and why this is important.

What are networks for?

In offices and schools, computers are usually connected to make a computer **network**.

Networks are complicated to set up. Networks are expensive to run because they need a trained person to fix problems. Most organisations think the cost is worth it because a network has many advantages.

The advantages of networks

- **Communicating:** We can send messages and emails over the network.
- **Sharing:** We use a network to share files and expensive devices like printers.
- **Saving work:** We can save files onto network storage drives. You can use your files on any computer on the network. It is easy to share files with other people.
- **Working together:** Networks help people to work together.

What makes a network?

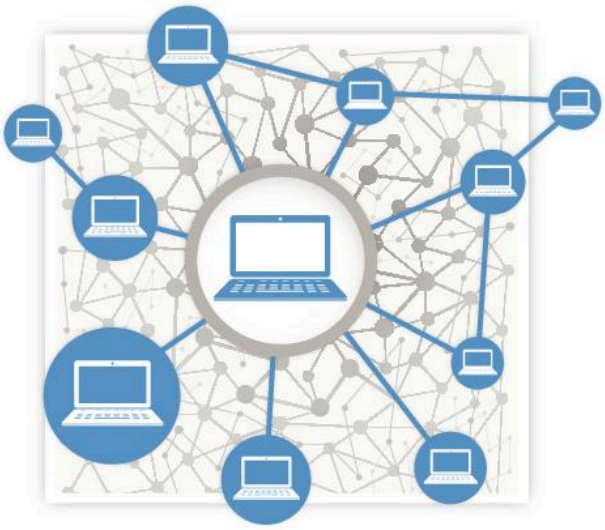
Four things are needed to make a network:

- **Network devices** to send messages and files between computers, and to store files and applications
- **Cables** to join the devices together
- **Network software** to make the devices work together
- Rules that allow all the parts of a network to work together

You will learn more about network devices and how they are connected in Lessons 1.2 and 1.3.



The two main types of network



- A **local area network** (LAN) joins computers in a single building. A LAN allows people in that building to work together. A school network is a LAN.
- A **wide area network** (WAN) joins computers that are far apart. An organisation with offices in different cities or countries uses a WAN so the employees can work together. The internet is a WAN.

The 'invisible' network

A school network can have 30 or more network devices. It can have 10 kilometres of cable running around the school.

A network is big, but most people don't notice it is there. Most of the equipment is locked away to keep it safe. But there are clues that the network exists. In this unit you will learn to spot the clues.



Activity

Web Valley Primary School has 60 network connections. The average cable used for each connection is 50 metres long. How many metres of network cable are there in the school? Convert your answer to kilometres.





Extra challenge

Use a spreadsheet to solve the problem in the activity.

Then use your spreadsheet to solve this problem:

Web Valley Secondary School has 140 network connections. The average cable is 65 metres long. How many kilometres of network cable are there in Web Valley Secondary School?

Think again

Do you save your school work to the school network? What is your personal area of the school network called? Is there an area on the network that your teacher uses to share files with the class?

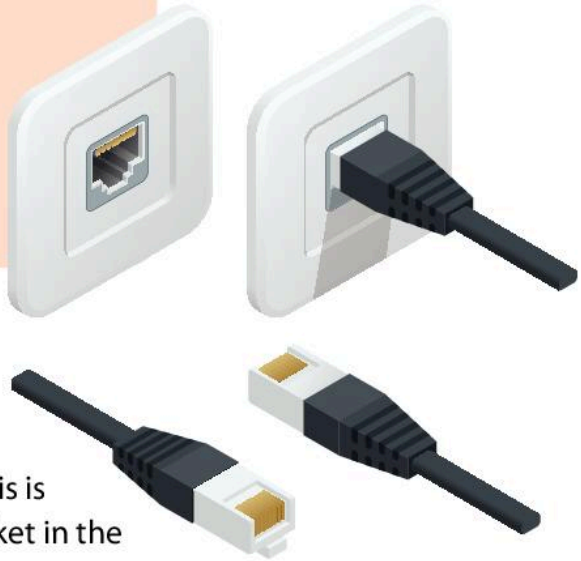
Where is the network equipment in your school stored?

1.2 Network connections

In this lesson

You will learn:

- how computers are connected to a network
- how to set a strong password and how to keep your password safe.



Connecting to a network

There are two ways to connect a computer to a network.

Wired connection

If you use a cable to plug the computer into the network, this is called a **wired connection**. A network cable connects a socket in the computer to an identical socket on the wall of a room.

If you see a socket like the one in the picture, it is a clue that the building you are in has a local area network (LAN).

Wireless (Wi-Fi) connection

You can also connect to a network using a **wireless connection**. This is also called **Wi-Fi**.

Networks use a device called a **wireless access point** (WAP) to provide a wireless connection. If you are close to a WAP, you can connect to a network without using a cable. A place where a wireless connection is available is called a hotspot.

You usually see WAPs high on the wall or on the ceiling.

Laptop and tablet computers usually use a wireless connection to a network. Desktop computers are usually connected to a network with a cable.



Using a network login

To use a network, you must have a **login**. A login is made up of a user name and a **password**.

Using a login keeps the network safe. Only people who have permission can use the network. Only you can see your own files on the network.



Keep your password safe

You must keep your password safe to protect your information and your work.

- Never tell anyone your password.
- Change your password regularly.
- Choose a strong password.

A strong password

A strong password is difficult for another person to guess. You can use the passphrase method to make a strong password.

1. Think of a short phrase that you can remember. Use two or three short words with eight to twelve characters in total. For example: **'Ginger cats'**
2. Remove the spaces between the words: **Gingercats**
3. Replace some of the letters with other characters that look similar. For example: a → @, e → 3, s → $.
4. This is your new strong password: **Ging3rc@t$**



Activity

Use the passphrase method to make a new strong password for your login.



Extra challenge

Create a poster that reminds students to keep their password safe. Include a guide for creating a strong password. Use your own example of a strong password. Can you suggest different characters to use instead of letters in the phrase?



Explore more

Logins are not just used for school networks. People use logins at work and for online shopping and banking. Ask an adult at home to help you make a list of all the things they use logins for.

Share the passphrase method of creating a strong password with the adult.

1.3 Network devices

In this lesson

You will learn:

- more about network devices, software and rules.

Devices in a network

A network needs special devices. Each device does an important job in the network.

Servers

A **server** is a powerful computer. A network uses several servers. Each server does a different job.

Print server: When you print a picture or a document, the print server makes sure your file is printed correctly.

File server: This server makes sure your documents are saved correctly.

Email server: This server makes sure your emails reach the right people.



Switches and hubs

Switches are like major junctions in a road traffic system. When you send a message, a switch decides which cable the data needs to travel down to reach the right destination. For example, if you send an email, the switch makes sure your email goes to the email server. A **hub** is a type of switch.

Routers

A **router** connects a network to the internet. Your school network has a router. If you have an internet connection at home, it uses a small router.



How do network devices work together?

The devices in a network work together to get a job done. Here is what happens if you send an email from your school to your home.

You press 'Send'.

Your computer sends the email to a **switch**.

The switch sends the email to the **email server**.

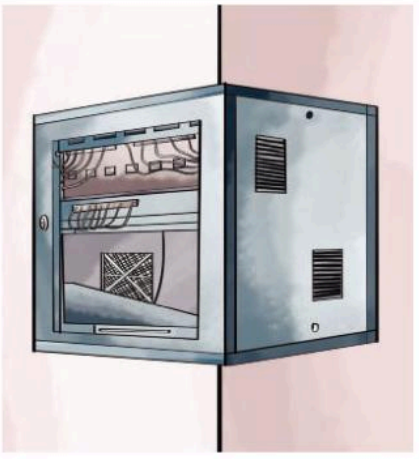
The email server sends the email to the **router**.

The router sends the email to the **internet**.

Your email reaches your home.

Where are network devices stored?

Network devices are usually stored in a locked room called a **server room**. They are stored in metal cabinets. Sometimes they are stored in small cabinets in classrooms.



Software and protocols

Two more things are needed to make a network work.

- Software is used to make sure the network runs properly. For example, network software is used to set up user logins.
- All parts of the network must understand each other. The whole network must follow rules so that jobs are done in the correct order. In a network the rules are called **protocols**.



Activity

Can you find any clues in your school that tell you the school has a network? If you can, take photos or make drawings as evidence.



Extra challenge

Read the description of what happens in a network when you send an email. Write a similar description of what happens when you print a document.



Explore more

If you have an internet connection at home, ask an adult to help you find the router. Be careful not to disturb any cables. Can you see anything that you recognise from studying this unit?

1.4 The internet

In this lesson

You will learn:

- what the internet is
- how the internet is different from the world wide web.

What is the internet?

The **internet** is a wide area network (WAN) that connects computers across the world. It is the biggest network in the world.

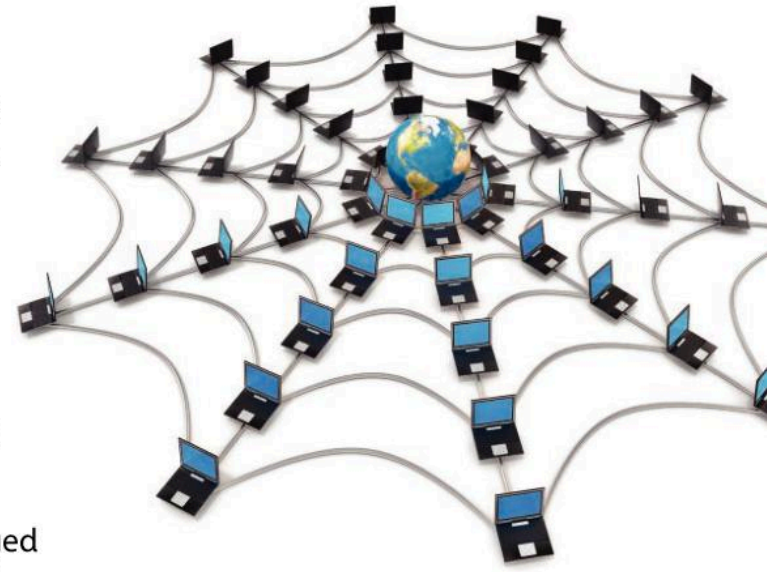
People use the internet to communicate and to share things with other people. The internet helps people to learn and work together.

The internet is not owned or controlled by a single person or organisation.

Internet services

The internet is a network of connections. Using those connections, we can make use of a range of services.

- **The world wide web:** Made of all the web pages in the world.
- **Email and chat:** Ways of sending messages and content to people you know.
- **Video conferencing:** People across the world can meet as if they were in the same room.
- **Streaming:** Games, music and videos can be played on your local device without being downloaded first.



The world wide web

The **world wide web** (www or just 'web') is a part of the internet. The web is all the websites and web pages stored on the internet. You use a web browser to look at web pages on the web. You use a search engine to search for information on the web.

The web is a powerful tool because every web page links to other pages. The links on web pages are called hyperlinks. If you drew lines between the hyperlinks on all the web pages on the web, it would look like a huge spider web.

Computers and the internet

You can use a computer to access all internet services. You use a browser to view the web. You can use other software applications (often called apps) for other internet services such as email.

Many people use a smartphone to access the internet. A smartphone can connect to the internet from anywhere that has a phone signal.

Smart devices

Many home devices are connected to the internet. These are called **smart devices**.

Home security systems can send a warning to your smartphone if sensors detect sound or movement in your home while you are out. You can check security cameras on your smartphone to see if there is an intruder.

You can use your smartphone to control home air conditioning and heating systems so the temperature is comfortable when you get home. You can switch on lights to make it look as if someone is home.





Activity

List the main internet services. Say which ones you have used. Give an example.



Extra challenge

Which devices do you use to connect to the internet? Which services do you prefer to access using a smartphone? Which services do you prefer to access using a computer? Give examples.

Think again

Research smart home heating or air conditioning systems on the web. List some of the features. How do smart home air conditioning and heating systems help the environment?

1.5 The changing world of work

In this lesson

You will learn:

- how computer communication has changed the way people work.



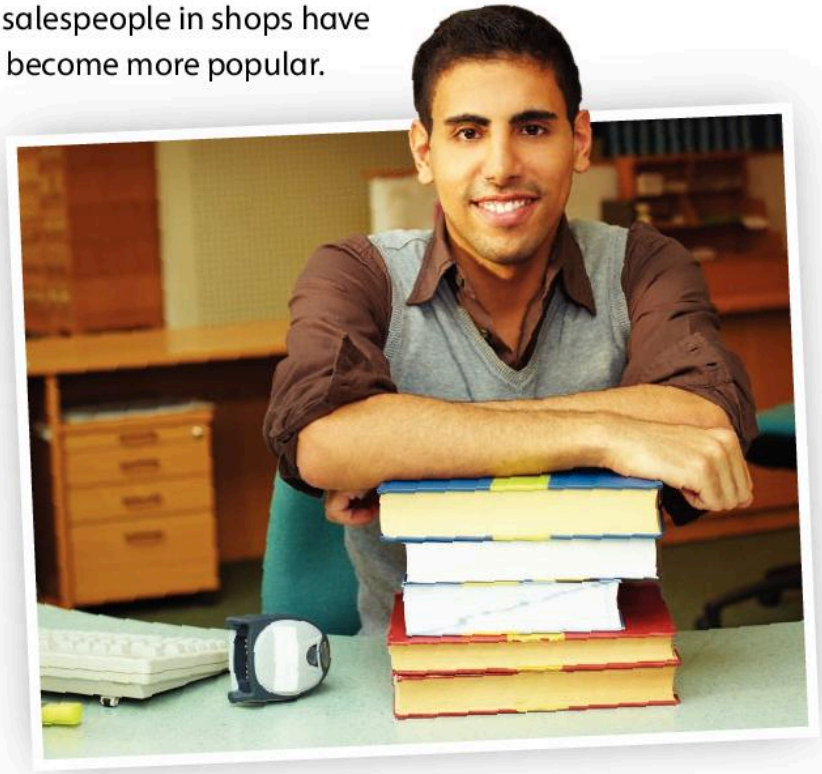
New jobs

The internet has changed the way people work. New jobs have appeared that did not exist before. There are technical jobs like network manager. There are creative jobs like web designer and games programmer.

Old jobs – new skills

The internet has changed existing jobs. A librarian used to work with books that were stored on shelves. Today, the librarian also needs the skills to find information on the web.

Some people have lost their jobs. Automation and robots have replaced low-skilled jobs in factories. Some salespeople in shops have lost their jobs because online shopping has become more popular.



As jobs change, people need to learn new skills. Some people find change exciting. Other people feel threatened by change.

The internet can make it easier to find a new job. There are websites that advertise job vacancies. People can apply for jobs by entering information about their skills on a database. Details of new job vacancies are emailed to them.

Different ways of working

Working in teams

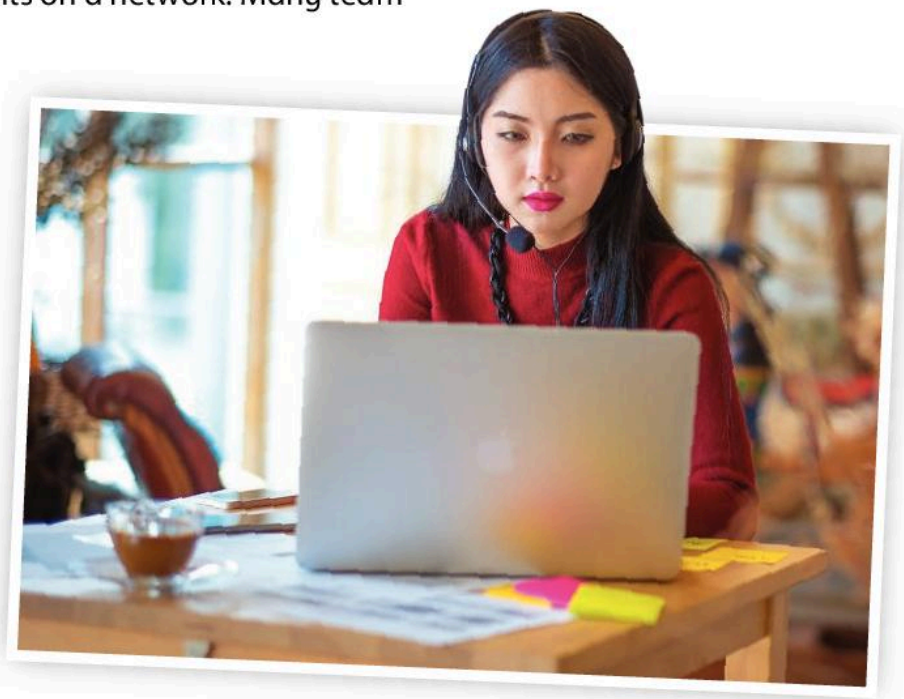
In the past, work teams mostly worked in the same building. The members of a team met regularly face to face to work on projects.

The internet allows people to communicate and share information with colleagues wherever they are in the world. Today, many teams are made up of people working in different cities and even countries. The team members share documents on a network. Many team members never meet face to face.

Teleworking

In the past, people travelled to an office or factory every day to do their work. The internet allows people to communicate using email, internet chat and video conferencing. People can access the information they need to do their work using networks and the internet.

This means that people can work from home. This is called **teleworking**. People don't have to travel so much and can fit work around their home life.





Activity

Describe your dream job when you grow up. How will you use a computer in your job?



Extra challenge

Make an advert for your dream job. Include the computer skills that are needed for the job.



Explore more

Find out what the adults in your family and other grown-up friends think about teleworking. Ask these questions.

1. Do you ever do part of your job from home using an internet link?
2. Could you do more of your job from home using an internet link?
3. Would you like to do more work from home?

Ask any other questions you can think of. Write down the people's answers so you can share them with the class.

1.6 Living with the internet

In this lesson

You will learn:

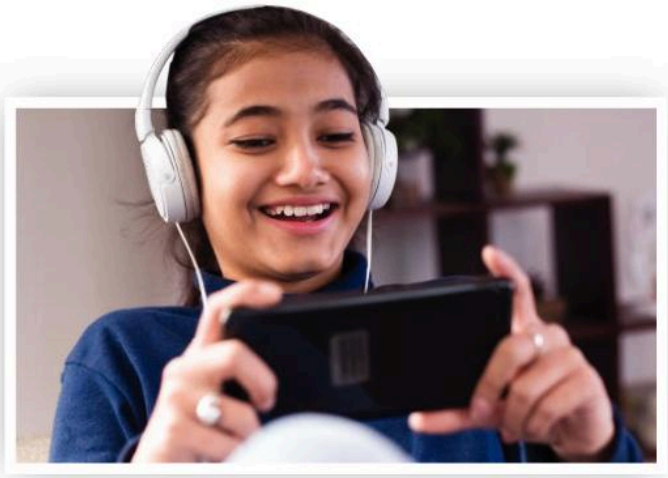
- how the internet affects our lives.

How the internet affects you

The internet has changed the way we lead our lives. Many of the changes have been positive.

Communication: The internet allows you to communicate using text, voice and video. You can share pictures and files. You can use the internet to find advice or to solve a problem.

Information: You use information every day to help you learn, work and lead your daily life. The internet has made information easier and quicker to find.



Convenience: The internet provides 'on-demand' services. This means you can use services when you want them. People use internet banking to pay for things they buy online. You can choose from thousands of shops on the internet. You can download music, games and videos instantly.

Remote control: In Lesson 1.4 you learned about smart devices in the home that allow you to control air conditioning, heating, lighting and security systems remotely.



The dark side of the web



If you use the internet responsibly, you will enjoy its benefits. But be aware that there is a negative side to the internet.

Not everything you read is true

The information on the web is often called content. Some of the content may be false. The writer may make a mistake or they may deliberately write false or misleading content. Some writers may write false content to persuade you to accept their opinions or to sell you something.

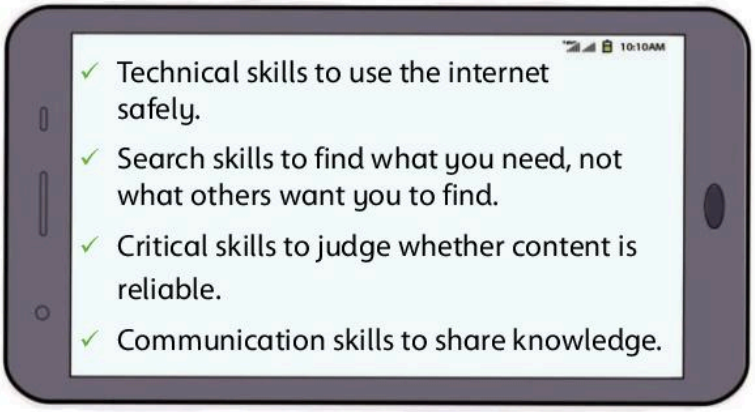
Don't trust everyone you meet online

Criminals use the internet. Some criminals use fake emails or websites to try to steal money or personal information. Other criminals trade illegal and counterfeit (fake) goods.

Some people use social media or text messages to send threats or to spread cruel rumours. If someone tries to bully or frighten you online, do not reply to the bully. Report the problem to a responsible adult.

Staying safe on the internet

To stay safe on the internet you need to develop these skills.



You will develop some of these skills in Unit 2.



Activity

Imagine that your friend has received threatening text messages from an anonymous person. Write a set of bullet points advising your friend how to deal with the problem.



Extra challenge

Create a poster with the title 'Good internet / Bad internet'. Use one half of the poster to promote the positive side of the internet. Use the other half to warn about the negative side of the internet.



Explore more

Find out how old your parents or grandparents were when the web was invented. Do they remember the first time they used the web?

Check what you know

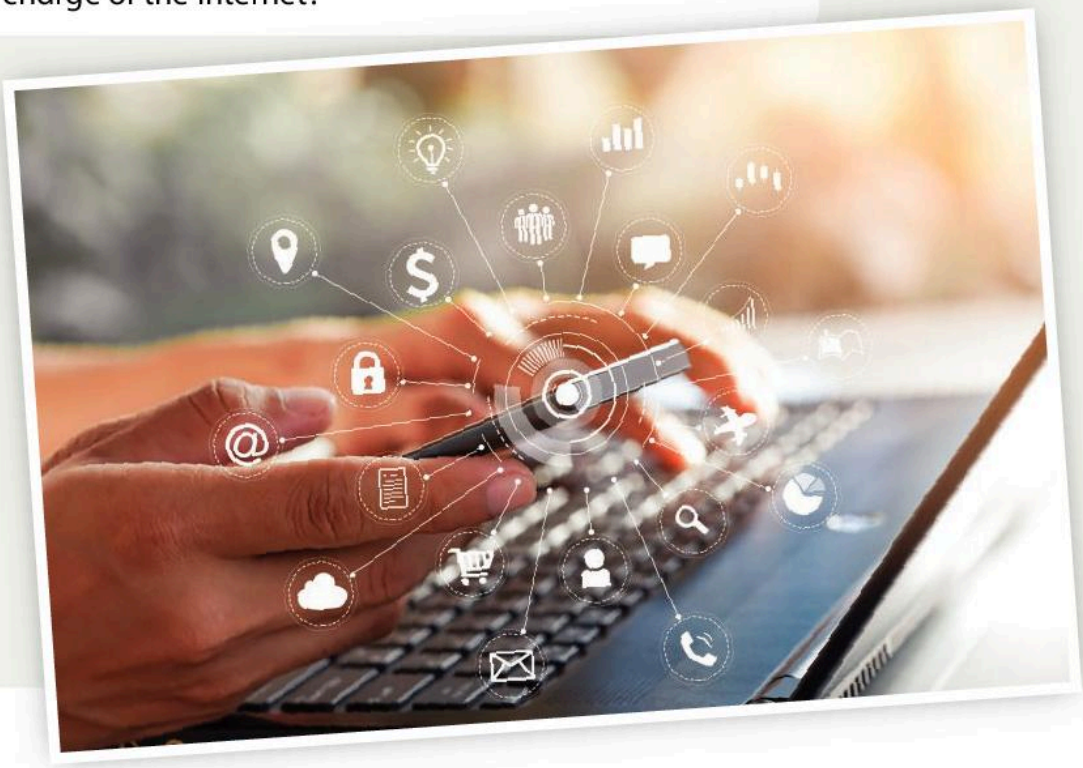
You have learned

- how computers and devices can be connected to make networks
- what the internet is and what services it provides
- how the internet helps us work together in the modern world.

Test

Think of a device you have used that is connected to the internet. You could choose your smartphone, your computer at school, or another device. Now answer questions about this device.

1. Name the device that is connected to the internet.
2. What is the internet?
3. How is the device connected to the internet – wired or wireless? Explain how you know.
4. When you look at the world wide web on your device, what do you see on the screen?
5. Name one other internet service you could use on the device.
6. Name one piece of equipment that helps your device connect to the internet. What does this equipment do?
7. Who is in charge of the internet?





Activity

Use a word processor to write a report about the internet. Use the headings shown. You can add images to the text if you want.

1 **Internet services:** Describe at least two services offered by the internet.

2 **Working world:** Describe how people might use the internet at work.

3 **Good or bad?** Discuss at least one good and one less good feature of the internet.

Self-evaluation

- I answered test questions 1 and 2.
- I completed section 1 of the activity. I described some positive and negative changes that the internet has caused.
- I answered test questions 1–5.
- I completed sections 1 and 2 of the activity. I described how the internet has changed the way that people work.
- I answered all the test questions.
- I completed all sections of the activity.

Re-read any parts of the unit you feel unsure about. Try the test and activity again – can you do more this time?